

Blood

Blood is a connective tissue in fluid form, red in color that delivers necessary substances to the cell.

Composition of blood:

Blood is made up of two parts as:

- **Cellular part** (formed elements) 45%.
 - i- RBC's (Red Blood Cells) or erythrocytes.
 - ii- WBC's (White Blood Cells) or leukocytes
 - iii- platelets
- **Non cellular** part of blood (plasma) 55%



Plasma:

A yellowish colored liquid that holds cellular part is called plasma.

Composition:

Plasma consists of a fluid part (water 91%) and a solid part 9% (soluble blood 7%, hormones, electrolytes & nutrients 2%).

Organic substances:

Proteins

- | | | |
|------|--------------------------|------------|
| i- | Albumin | 4.5 – 5.5% |
| ii- | Fibrinogen | 0.2% |
| iii- | Globulin | 1.3 – 2% |
| iv- | Prothrombin | 0.1% |
| v- | Plasma complement system | |

Inorganic substances:

Sodium, potassium, chloride, calcium, bicarbonate, magnesium and phosphate.

Properties of blood:

Color:

Blood is red in color. Blood contain O_2 is bright red color, while CO_2 containing blood of purple reddish color.

Volume:

- Male => 5.0 Liters
- Female => 4.7 Liters
- New born => 450 ml

pH:

Blood is slightly alkaline and its pH in normal condition is 7.4.

Viscosity:

Blood is 5 times more viscous than water. It is due red blood cells and plasma proteins.

Cellular contents of blood

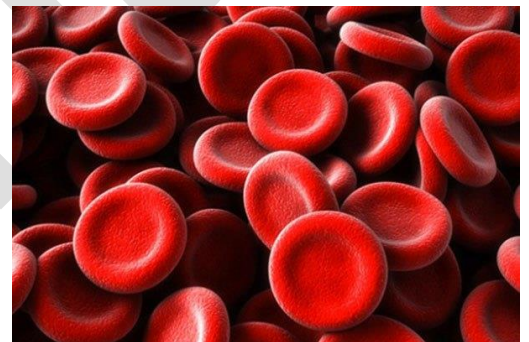
- 1- **Red Blood Cell** (RBC's or Erythrocytes)
- 2- **White Blood Cell** (WBC's or Leukocytes)
- 3- **Platelets** (Thrombocytes)

Red Blood Cells

RBC's are non-nucleated, oxygen carrying hemoglobin containing cells present in blood.

Properties:

- i- Round
- ii- Small
- iii- Biconcave
- iv- Life span is 120 days (100-120)
- v- Non-nucleated
- vi- No DNA
- vii- No Mitochondria and Golgi apparatus



Production:

Erythropoiesis: (ERYTHROPOITEN)

Erythropoiesis is a process in which RBC's are produced from stem cells in a bone marrow to mature RBC's within 7 days using hormone ERYTHROPOITEN.

(Tibia, Fibula and Ribs)

Apoptosis/ERYPTOSIS:

Programmed cell death of RBC's is called ERYPTOSIS.

Count:

Male: 5.4 million/ mm³

Female: 4.7 million/ mm³

Infants: 6.0 million/ mm³

Composition of RBC's:

Mature RBC's do not contain mitochondria and nucleus. They contain Hb.

Water = 65%

Solid and semi-solid = 35%

HB = 33%

Organic and inorganic substances = 2%

Proteins, phosphate, cholesterol, urea, creatinine, amino acid.

RBC's Increase value:

- After meals
- After exercise
- At high temperature
- At high altitude

RBC's decrease value:

- During pregnancy
- Repeated X-ray
- During sleep

Functions of RBC's:

1- Respiration:

RBC's contain hemoglobin which transports oxygen and carbon dioxide. It also contains carbonic anhydrase enzymes which accelerates the following reaction.



This helps to transport CO₂ from tissue to the lungs in HCO₃ form.

2- Acid base balance:

RBC's help to maintain acid base balance by buffering action of hemoglobin and other intracellular buffer.

3- Maintain ionic balance:

RBC's maintain ionic balance by special permeability of cell membrane.

4- Viscosity of blood:

RBC's help to maintain the viscosity of blood.

Hemoglobin

Hemoglobin is a red, oxygen carrying pigment present in RBC's. It consists of 4% heme and globin 96%.

Or

Iron containing biomolecule that binds oxygen and is responsible for the red color of cells and bones.

Normal value of hemoglobin:

Male = 14-16gm/100ml

Female = 12-14gm/100ml

New born = 23gm/ 100ml

Regulation of Hb:

Regulation of Hb depends upon the count of RBC's so its regulation depends upon the RBC's regulation.

Physiological variations:

Any condition that increase or decrease the RBC's count will effect Hemoglobin concentration.

Functions of hemoglobin:

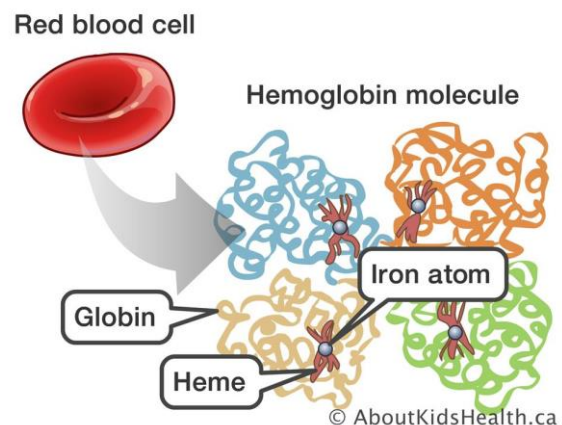
Oxygen transport:

It is essential for O₂ carriage; it combines reversibly and loosely with oxygen and supplies tissues from the lungs in the form of oxyhemoglobin.

Carbon dioxide transport:

Carry same amount of carbon dioxide to the lungs from the tissues.

Acid – base balance:



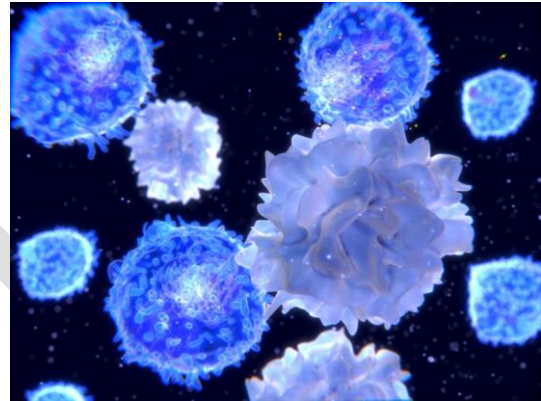
Due to the globin fraction it acts as a strong buffer and helps to maintain the pH of the blood.

Pigments:

Different pigments of bile, stool, urine etc are formed from hemoglobin and excreted through the urine and stool.

White Blood Cells

White blood cells are colorless and nucleated elements of blood involved in the immunity/defense system of body.



Properties:

- 1- Colorless
- 2- Nucleated
- 3- Mobile unit of bodies defense system
- 4- Size is comparatively larger
- 5- Short life span 13-20 days
- 6- Irregular shape

Count of WBC's:

7000/ mm³ (Average)

4000-11000/ mm³ (Range)

- Produce in bone marrow.
- Destroyed in the Lymphatic system.

Classification of WBC's

They are classified into two major groups:

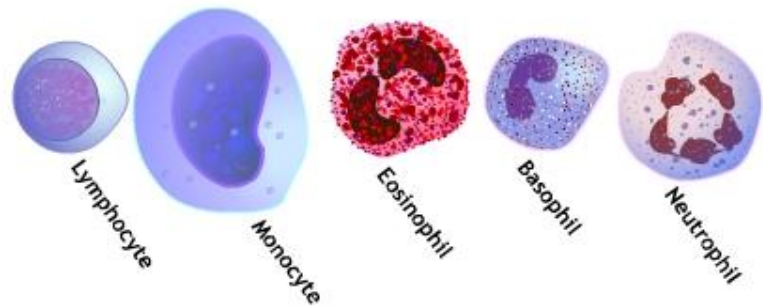
- 1- Granulocytes
- 2- Agranulocytes

Granulocytes:

- These are WBC's with poly-lobed nucleus which are characterized by the presence of granules in their cytoplasm.
- 65% of total WBC's.
- Granules are present.
- Produced in bone marrow.

Cell types of Granulocytes:

- Neutrophils (62.0%)
- Eosinophils (2.3%)
- Basophils (0.4%)



Neutrophils (Phagocyte):

It is the type of WBC's that help in fighting against by ingesting microorganisms. (phagocytose bacteria and virus).

Eosinophils (Detoxification):

These are a type of WBC's that fight against parasitic infections.

Basophils (Allergic):

Are a type of WBC's that play their defensive role in allergic conditions and imported role in healing.

Agranulocytes:

- These are WBC's with one-lobed nucleus which are characterized by the absence of granules in their cytoplasm.
- 35% of WBC's.
- One-lobed nucleus.
- Absence of granules.
- Produced in Lymphoid tissue.

Cell types of Agranulocytes:

- 1- Lymphocytes (30%)
- 2- Monocytes (5.3%)

Lymphocytes:

Present in => Lymph

Lymphocytes are a type of WBC's that help in defensive mechanism of body by providing cell mediated and hormonal mediated immunity through killer cells, β -cells and T- cells.

Monocytes:

Are a type of WBC's important to regulate immunity against foreign substances and facilitate healing and repair.

Normal value of each type of WBC's

Neutrophils 5400/ mm³

Eosinophils 275/ mm³

Basophils 35/ mm³

Lymphocytes 2750/ mm³

Monocytes 450/ mm³

Platelets (Thrombocytes)

A small colorless disc shaped cell without a nucleus found in large numbers in blood and involved in clotting.

Properties:

Shapes:

These are small, irregular, round, and oval.

Size of platelets:

2-4 micron in diameter.

Count:

Their normal count in blood between
150,000 to 300,000/ mm³

Life span:

4-9 days

Ratio:

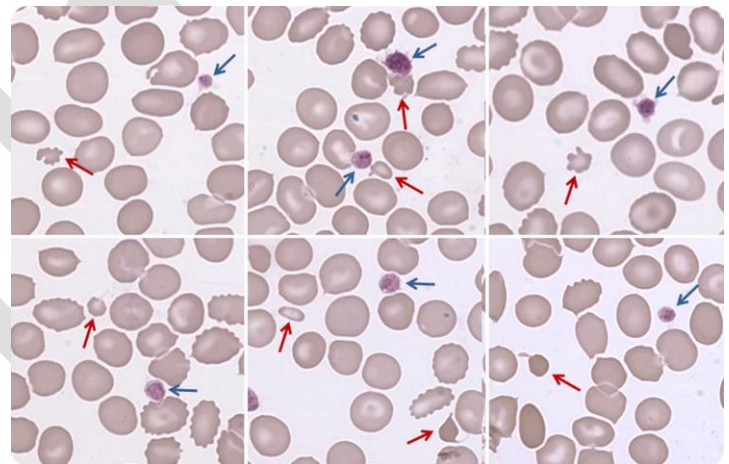
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Types:

- Small (young) platelets (or active platelets).
- Large (old) platelets.

Chemistry:

Platelets possess protein, lipids (phospholipid, arachidonic acid).



Adhesion:

Attachment of platelets to interrupted endothelium.

Activation:

Platelets change their shape, turn on receptors and secrete chemical messengers.

Aggregation:

Platelets connect to each other through receptor bridges.

Types of platelets as:

1- Thrombocytopenia:

Level of platelets decreased.

2- Thrombocytosis:

Level of platelets increased.

3- Thrombosis

Formation of blood clot in veins and arteries.

Function of platelets**Hemostasis:**

- i- Platelets adhesion
- ii- Clotting of blood
- iii- Platelets aggregation
- iv- Platelets released reaction

Other function of platelets:

- i- Platelets are necessary for maintenance of vascular integrity.
- ii- Platelets repair small vascular injuries by adhering to basement membranes.
- iii- Platelets provide glycoprotein, their adhesion to collagen.
- iv- Platelets show slight phagocytic activity.

Indications for platelets transfusion:

- i- Bleeding and Thrombocytopenia.
- ii- Platelets below 40,000.
- iii- Platelets count dysfunction.

FUNCTIONS

- Role in haemostasis.
- Role in clot formation.
- Role in clot retraction.
- Role in repair of injured blood vessels.
- Role in Defense mechanism.
- Transport & storage function.

- iv- Acute disseminated intravascular coagulations.

Blood groups

Blood group is the classification of blood based on the presence or absence of antigenic substances on the surface of RBC's.

Types:

OAB or ABO blood groups [classical blood groups] ABO blood groups are further classified into:

- 1- O = 47%
- 2- A = 41%
- 3- B = 9%
- 4- AB = 3%

Blood Group	Gives to these groups	Receives from these groups
	All	O- only
	AB+, A+, B+, O+	O- and O+
	AB-, AB+, A+, A-	O- and A-
	AB+ and A+	O-, O+, A-, A+
	B-, B+, AB-, AB+	O- and B-
	B+ and AB+	O-, O+, B-, B+
	AB- and AB+	O-, A-, B-, AB-
	AB+ only	All

Terminology Related To Blood Groups

Antigen:

An antigen is any substance that cause your immune system to produce antibodies. For example: Bacteria, virus, chemicals.

Or antigen are foreign substances that produce immunological response when they enters the body.

Agglutinogen:

Any antigen (antigen-A or antigen-B) present in blood and activates immune system to produce antibodies.

Antibody:

A specific substance or protein produce in response to antigens is known as antibody.

Or these are the substances produced in response to antigen.

Agglutinin:

A substance or protein like antibody produce in response to antigen.

Agglutination:

The clumping of RBC's when antibodies bind to specific antigens.

Rh factor:

Rh factor are the antigens present on the surface of RBC's which causes agglutination on mixing with anti-Rh-D-serum.

Rh blood group system:

a- Rh +ve

b- Rh -ve

Genotype:

This is the combination of different genes.

Phenotype:

This is the way in which the individual gene expresses itself.

On surface is Rh +

No surface is Rh-

Genotype	Phenotype	Allele A	Allele B
DD	Rh +	D	D
Dd	Rh +	D	d
dd	Rh -	d	d